FROM ROBOTS TO COBOTS

Machines · Robots · Collaborative Systems

An educational journey into intelligent automation

OVERVIEW

What We Will Explore

01 **Machines**
Purpose-built devices with predefined operations

02 **Robots**
Complex machines that sense, decide & act

03 **Robot Anatomy**
Sensors · Actuators · Controllers · Structure

04 **Cobots**
Collaborative robots designed to work with people

05 **Key Differences**
How machines, robots and cobots relate

MACHINES

What is a Machine?

- Designed to perform specific, predefined tasks
- Operations are fixed by mechanical, electrical or software design
- Limited or no ability to sense and adapt to environment
- Requires human control or simple programming for each cycle



Example: Washing Machine

EXAMPLE

Washing Machine — A Classic Machine

Purpose

Designed specifically to clean clothes. One primary function.

Control

Controlled by a small embedded program with fixed wash cycles.

Limitation

Cannot sense dirt level or adapt recipe. No environmental awareness.

ROBOTS

What Makes a Robot?

A complex machine capable of sensing its environment, making decisions, and performing a variety of tasks — often with partial or full autonomy.

01	**Sense**	Uses sensors to perceive the world
02	**Decide**	Processes data to choose actions
03	**Act**	Moves and manipulates via actuators
04	**Adapt**	Can learn and update behaviour



APPLICATIONS

Where Robots Excel

Manufacturing

Welding, assembly, material handling, painting

Healthcare

Complex medical surgeries, rehabilitation, logistics

Defence & Security

Surveillance, bomb disposal, reconnaissance

Exploration

Space, underwater, hazardous environments

ANATOMY

Basic Structure of a Robot

01

Sensors

Eyes, ears & touch — gather data from the environment

02

Actuators

Muscles — motors, gears, hydraulics that create motion

03

Processors

Brain — compute decisions from sensor data

04

Outer Body

Skeleton & skin — structure, protection and appearance

HOW IT WORKS

The Sense → Think → Act Loop

SENSE	THINK	ACT	LEARN
Sensors collect data about the environment and the robot itself	Controller processes inputs and decides the next action	Actuators execute the motion or task	Update models / behaviour for better future performance

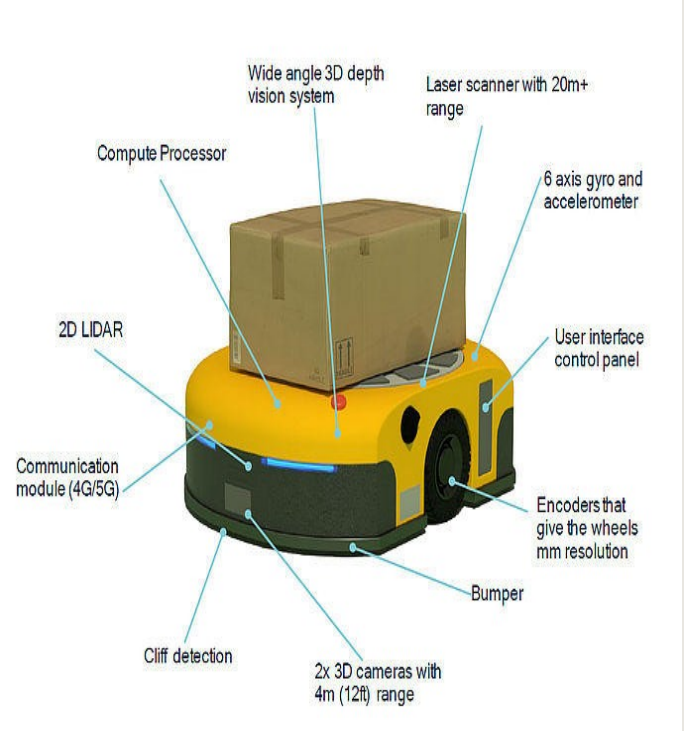
Robots continuously close this loop to interact intelligently with the world.

SENSORS

Sensors — The Robot’s Senses

Enable the robot to learn about its environment, move safely, avoid obstacles and interact with objects.

- Cameras (vision)
- LiDAR / Lasers
- Ultrasonic
- Inertial (IMU)
- Force / Tactile
- Temperature
- Microphones
- Encoders



ACTUATORS

Actuators — Creating Motion

Actuators are the muscles of the robot. They convert electrical signals into physical movement.

Electric Motors

Most common — DC, servo, stepper motors for precise positioning

Gears & Transmissions

Reduce speed, increase torque, change direction of motion

Hydraulics / Pneumatics

High force applications (heavy lifting, industrial presses)

Linear Actuators

Push / pull motion for grippers, lifts and linear joints

CONTROLLERS

Controllers — The Robot’s Brain

- Receives data from all sensors
- Runs algorithms / AI models
- Sends commands to actuators
- Works from pre-written programs or learned policies
- Can be embedded (on-robot) or remote (cloud)

KEY INSIGHT

The controller turns raw sensor numbers into intelligent behaviour.

Without it, a robot is just a pile of motors and cameras.

EXAMPLES

Example: Surveillance Robot Dogs



Mobility

Walks on four legs over complex terrain

Sensing

Cameras, LiDAR, microphones for 360° awareness

Autonomy

Can patrol, follow paths, detect anomalies

Role

Acts as a mobile watchman for security & inspection

COMPARISON

Machines vs Robots

MACHINES

- Single / few fixed tasks
- Predefined operations only
- Little or no sensing
- No decision making
- Human or simple program control
- Example: Washing machine, printer

ROBOTS

- Multiple / complex tasks
- Can adapt behaviour
- Rich sensor suite
- Autonomous decision making
- Sense → Think → Act loop
- Example: Robot arm, Spot dog

COBOTS

Collaborative Robots (Cobots)

Cobots are robots specifically designed to work safely alongside humans, sharing the same workspace and tasks.

Human collaboration	Designed for direct interaction with people
Safety first	Force-limited, soft materials, collision detection
Easy programming	Often teach-by-demonstration, no specialist code
Flexible deployment	Quick to redeploy for different tasks



COMPARISON

Traditional Robots vs Cobots

Aspect	Traditional Robot	Cobot
Workspace	Caged / separated	Shared with humans
Safety	Physical barriers	Inherent (force, speed limits)
Programming	Complex, specialist	Intuitive, teachable
Flexibility	High volume, fixed	High mix, easy re-task
Focus	Full autonomy	Collaboration & assistance

IMPACT

Why Cobots Are Important

Safer workplaces

No cages needed; force sensing prevents injury

Human + Robot teams

Best of both: human creativity + robot precision

Faster ROI

Easier to deploy and reprogram for SMEs

Higher productivity

Humans focus on high-value work, robots on repetitive



SYSTEMS

Multi-Robot Systems

Multiple robots working together (and often with humans) over a network to achieve a common goal.

1

Smart Home / Office

Cleaning robots, security bots and delivery bots coordinating via the cloud

2

Firefighting

Teams of ground and aerial robots mapping fire, delivering water, rescuing people

3

Warehouse / Factory

Fleets of AMRs and cobots sharing maps and tasks in real time

SUMMARY

Key Takeaways

1	**Machines**	Do one (or few) predefined jobs extremely well.
2	**Robots**	Sense, decide and act — they are machines that can adapt.
3	**Anatomy**	Sensors + Actuators + Controllers + Body form the core.
4	**Cobots**	The bridge: robots that safely share space and work with people.
5	**Future**	Multi-robot + human teams will become the new normal.

Thank You

From Machines → Robots → Cobots
The journey of intelligent automation continues

Questions & Discussion